

RTL Development on CAE

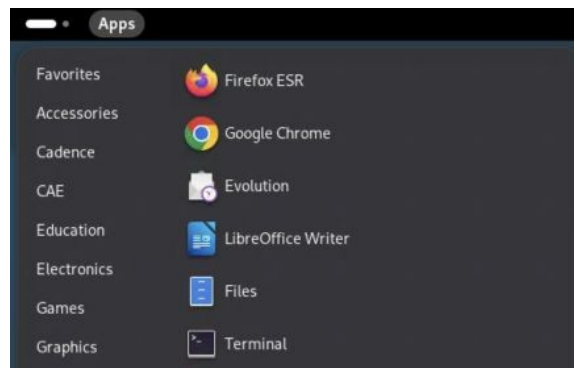
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Use this guide to inspect and simulate RTL, debug its behavior in Verdi, run a synthesis sanity check, and inspect the resulting schematic in Design Vision. The small register makes the entire loop observable before you apply it to a compute, control, or memory block.

Sign in to Guacamole

Guacamole displays a CAE Linux desktop in your browser. The tools run on the CAE machine. Use your own UW NetID and confirm that your account can reach CAE Linux services before the lab.

1. On your laptop, open <https://guacamole.cae.wisc.edu> in a browser.
2. Complete the UW web sign-in with your NetID and password, then complete MFA if requested.
3. At the Linux login screen, enter your UW NetID and password again. Use your NetID credentials at this second prompt too. The old separate CAE username and password are not the instructions for this service.
4. Wait for the Linux desktop. Open Apps in the upper-left corner and choose Terminal under Favorites. Maximize the terminal by double-clicking its title bar.
5. Keep the Guacamole tab open. A terminal inside this desktop is already on CAE; do not SSH back into CAE from it.



Open Terminal from the CAE Apps menu.

If you already have an active session, the browser may reconnect without showing every login screen. If sign-in fails, record which prompt failed: UW web sign-in, MFA, or the Linux desktop login. These are different stages.

Official CAE login instructions and screenshots: <https://kb.wisc.edu/cae/163323>

Enter the Synopsys environment

Run these commands in the CAE terminal, one line at a time. Do not type a prompt such as \$ or synopsys> before a command.

```
module load synopsys/suite
synopsys-run
command -v vcs dc_shell verdi icc2_shell pt_shell
```

The prompt becomes synopsys>. The module selects the site setup; the container provides the operating environment expected by the tools. Load the module before entering the container and repeat this setup for every new terminal used for Synopsys tools.

```
synopsys> command -v vcs dc_shell verdi icc2_shell pt_shell
/srv/auto/apps/synopsys-2026/vcs/Y-2026.03/bin/vcs
/srv/auto/apps/synopsys-2026/syn/Y-2026.03/bin/dc_shell
/srv/auto/apps/synopsys-2026/verdi/Y-2026.03/bin/verdi
/srv/auto/apps/synopsys-2026/icc2/X-2025.06-SP3/bin/icc2_shell
/srv/auto/apps/synopsys-2026/prime/Y-2026.03/bin/pt_shell
synopsys>
```

Tool paths inside the working CAE Synopsys container.

Get the companion exercise onto CAE

On your laptop, open Terminal or PowerShell. Clone the guide repository, then copy it to your CAE home directory:

```
git clone https://github.com/abhinavnandwani/cae-synopsys-guides.git
scp -r cae-synopsys-guides YOUR_NETID@best-tux.cae.wisc.edu:~/
```

Replace YOUR_NETID with your NetID. Check a new SSH host key against the CAE fingerprint at <https://kb.wisc.edu/moodle/162780> before accepting. If Git is unavailable, extract the repository ZIP, rename its folder to cae-synopsys-guides, then run the SCP line. Guacamole does not provide file transfer. The repository README also explains cloning directly on CAE.

Back in the CAE container terminal:

```
cd ~/cae-synopsys-guides/lab
pwd
ls
```

You should see run_lab.py, rtl, tb, synth, and README.md. The recorded screenshots use a dated demonstration folder. Use the folder you copied. Keep work in your home directory so it is available across CAE hosts.

Work confidently in the terminal

The current directory affects relative paths. `pwd` prints it, `ls` lists files, `cd ..` moves to the parent, and `cd ~/cae-synopsys-guides/lab` returns to the exercise. A leading `/` means an absolute path; `~` means your home directory. Quote paths containing spaces.

Prompt or location	Commands that belong there
Laptop Terminal or PowerShell	<code>ssh</code> and <code>scp</code> to reach CAE or transfer files.
CAE host shell	<code>module load synopsys/suite</code> , then <code>synopsys-run</code> .
Container <code>synopsys></code>	Linux commands, <code>python3 run_lab.py</code> , <code>vcs</code> , <code>verdi</code> , <code>dc_shell</code> , and <code>icc2_shell</code> .
Tool prompt such as <code>dc_shell></code>	Tool Tcl commands such as <code>read_ddc</code> , <code>report_timing</code> , and <code>help</code> .

Use these from the exercise folder to inspect source and logs without changing them:

```
ls -lh
cat rtl/sb_flop.v
sed -n '1,80p' tb/tb.sv
less runs/YOUR_RUN/compile.log
grep -nE 'Error|Fatal|Warning' runs/YOUR_RUN/compile.log
tail -n 30 runs/YOUR_RUN/simulate.log
```

Replace `YOUR_RUN` with an actual directory printed by the runner. In `less`, use Space to advance, `/Error` then Enter to search, `n` for the next match, and `q` to return to the shell. An Up-arrow recalls a command; Tab completes a path. Read a command before rerunning it.

For an interactive foreground command that is stuck, `Ctrl+C` requests interruption. For a GUI launched with `&`, close the application through its File menu. `jobs` lists jobs started by that shell. Keep logs: `command > run.log 2>&1` sends both standard output and errors to a file. Immediately after a command, `echo $?` reports its exit status, but a zero status alone does not prove the design passed.

Connect without the browser when useful

For terminal-only work, open a terminal on your laptop and run `ssh YOUR_NETID@best-tux.cae.wisc.edu`, then load the module and enter the container as above. Use Guacamole for the GUI steps in this handout. SSH and Guacamole may reach different hosts, but your CAE home directory is shared.

When finished, save work, close the EDA applications, and run `exit` to leave the container. Log out of the Linux desktop from the upper-right system menu. Closing the browser alone can leave the session running during CAE's two-hour reconnection window.

Read the RTL as a hardware contract

From the exercise directory, inspect both the design and the test that drives it:

```
cat rtl/sb_flop.v
sed -n '1,100p' tb/tb.sv
```

```
synopsys> cat ../../rtl/sb_flop.v
module sb_flop(input clk, input rst, input d, output reg q);
  always @(posedge clk)
    if (rst) q <= 1'b0;
    else    q <= d;
endmodule
synopsys>
```

The source of the one-bit register in the CAE terminal.

The top module is `sb_flop`. Its inputs are `clk`, `rst`, and `d`; its output is `q`. The `always` block runs on a rising clock edge. Reset is active high and synchronous: asserting it between edges does not immediately update `q`. The nonblocking assignment schedules the register update for that simulation time step.

Read the testbench next. `tb` instantiates the design as `dut`, generates the clock, and checks values after clock edges. The design top and the simulation top are different: synthesis uses `sb_flop`, while VCS uses `tb` for this test. Selecting the wrong top can produce an apparently successful build that never applies your stimulus.

Organize source changes before building

Keep synthesizable design source in `rtl/`, testbench code in `tb/`, and tool scripts in their own directories. Generated netlists, databases, logs, and waveforms belong in run directories. Edit the design source, then rebuild; editing `mapped.v` does not fix the RTL that generated it.

For your real block, write down port names, widths, clock and reset behavior, transaction acceptance, output latency, and backpressure or completion rules before relying on a test. Mark unresolved interface decisions explicitly. The architecture repository is the shared source for the top-level diagram and interface decisions; Compute 1 and Compute 2 continue their independent investigations.

Compile and simulate from the terminal

Use the verified runner for the first iteration:

```
cd ~/cae-synopsys-guides/lab
python3 run_lab.py simulation
```

Find the printed RUN_DIRECTORY. Inspect compile.log first, then simulate.log. Confirm the SB_SNPS_SIM_PASS marker and the expected run verdict. The runner also verifies that the deliberate fault is caught. It does not interpret raw exit zero as sufficient evidence of success.

For a larger block, keep a source manifest that lists the exact package, include, RTL, and testbench files. Package definitions must be available before code that imports them. Keep parameter overrides and the selected top in the build command or script, not in someone's terminal history.

The core compile and run pattern for this example is:

```
vcs -full64 -sverilog -timescale=1ns/1ps \
  -ntb_opts uvm-1.2 -debug_access+all -kdb \
  -cm line+cond+tgl+branch+assert \
  /absolute/path/to/rtl/sb_flop.v \
  /absolute/path/to/tb/tb.sv -top tb -o simv
./simv -cm line+cond+tgl+branch+assert
```

Run manual builds in a fresh run directory, with VERDI_HOME set as in run_lab.py. Replace the example absolute paths. The runner already does this setup, saves logs, and enforces a timeout. The direct commands show what it is doing and what must change for your block.

Diagnose the first failure

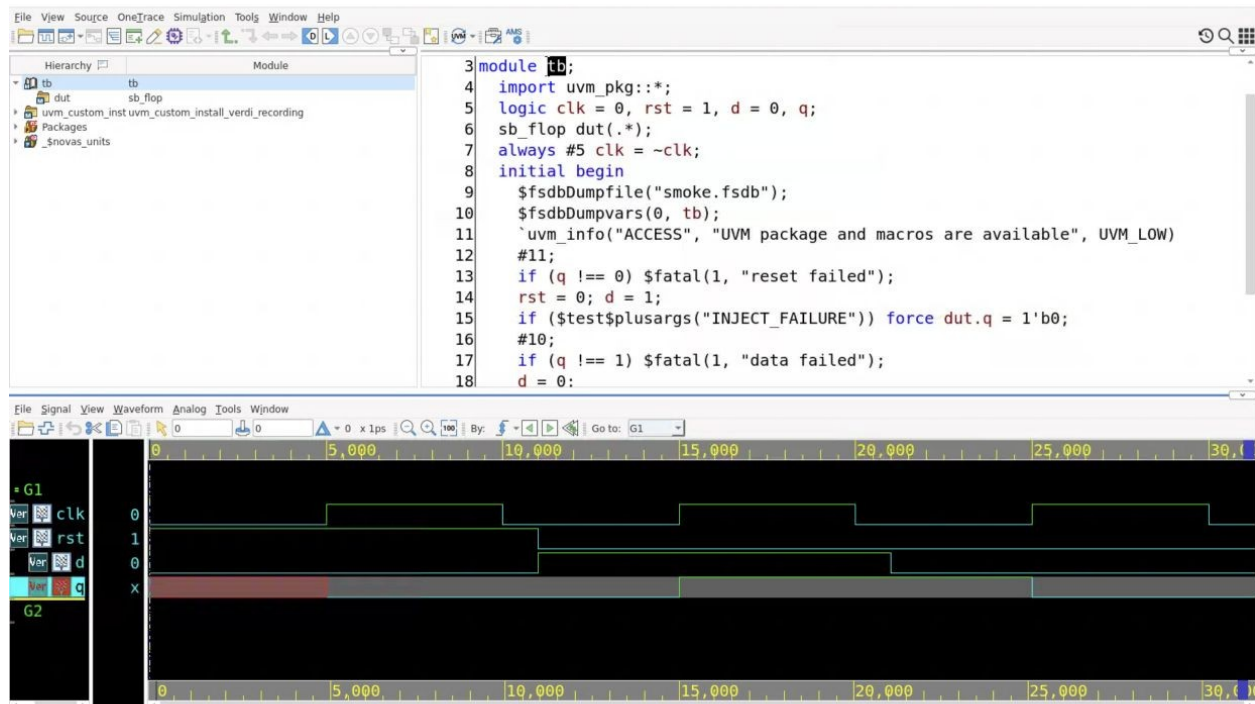
Symptom	First inspection
Missing module or package	Check the file manifest, include paths, package order, and selected top.
Width or signedness warning	Compare the source declaration with the interface contract and expression widths.
Unknown output	Check reset, initialization, driving logic, and whether the output has been sampled after an active edge.
Correct compile but wrong test result	Inspect stimulus, reset polarity, cycle timing, and the checker in the waveform.
GUI shows old source	Rebuild and open the waveform and debug data from the same new run directory.

Treat warnings as questions to resolve. A successful compile does not establish protocol behavior, numeric accuracy, clock-domain safety, or complete test coverage.

Trace source and behavior in Verdi

Change into the passing simulation directory and launch:

```
verdi -ssf smoke.fsd
```



The source and waveform views used to connect RTL statements to signal behavior.

1. Maximize the window. In the hierarchy, select tb, then expand it and select dut. Confirm that the source pane shows sb_flop.
2. Identify the edge-sensitive block and reset condition. Return to tb and identify when the stimulus changes relative to the clock.
3. In nWave, open Signal, Get Signals. Select /tb, add clk, rst, d, and q with Apply, then click OK.
4. Choose nWave View, Zoom, Zoom All. Place the cursor near a rising edge and compare all four signals at the same time.
5. Follow reset through the first edge at 5 ns. Then inspect the input change at 11 ns and output change at 15 ns. Explain the delay in terms of the source, not just the shape of the waveform.
6. Inspect the 21 ns and 25 ns transitions. If an output differs from the test expectation, find the first differing time and work backward through its inputs and driving statement.

Do not infer reset behavior from a signal name alone. This example uses synchronous reset. Your block's reset polarity, synchronization, and release requirements must come from its interface contract. Use the appropriate stimulus and sampling convention when you adapt the test.

If signals are missing, confirm the selected scope and dump coverage. If a value column looks surprising, check the cursor time. A value at the initial cursor is not the final value in the simulation.

Run a synthesis sanity check

From the exercise root, run:

```
python3 run_lab.py synthesis
```

The runner selects the teaching timing library, sets the source path, and calls Design Compiler with `synth/run.tcl`. It checks the log, completion marker, and generated outputs. Inspect the new run's `synthesis.log`, `mapped.v`, `area.rpt`, `timing.rpt`, and `constraints.sdc`.

The script reads RTL, selects `sb_flop`, links references, defines the clock and I/O timing, compiles, checks the design, and exports a mapped database and netlist. The 10 ns clock and 1 ns interface delays are teaching assumptions. Your real block needs constraints agreed with its surrounding interfaces.

Open the mapped design in Design Vision

In Guacamole, change into the printed synthesis run directory and run `design_vision`. Allow startup to complete, then maximize the window. The bottom input line is a Tcl command prompt, not a Linux shell. Enter the following there, one command at a time:

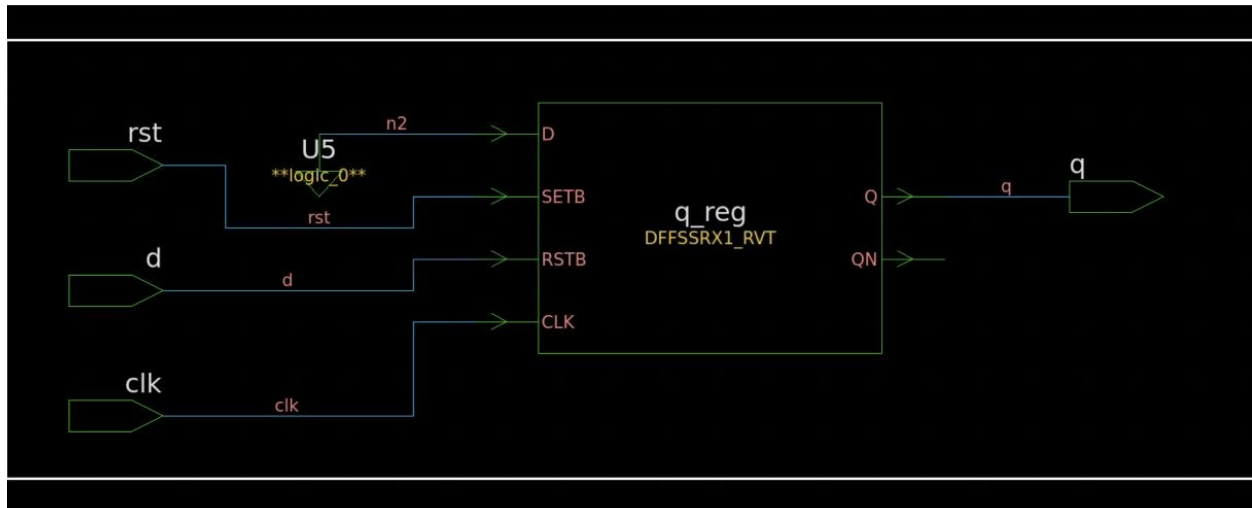
```
set edk /srv/auto/apps/saed32_edk/2023/lib/stdcell_rvt
set dbdir "$edk/SAED32_EDK/lib/stdcell_rvt/db_nldm"
set lib "$dbdir/saed32rvt_tt1p05v25c.db"
set_app_var target_library [list $lib]
set_app_var link_library [concat * $target_library]
read_ddc mapped.ddc
current_design sb_flop
link
read_sdc constraints.sdc
check_design
```

Inspect the console after every command. The expected current design is `sb_flop`, and references must resolve. If a library cannot be found, check the path and environment before interpreting the schematic. Loading a DDC and seeing a top module are not sufficient if linking still fails.

The DDC is a tool database for resuming inspection. Keep RTL, scripts, and constraints as the maintained inputs. Preserve the tool version with any saved database.

Inspect the schematic in the GUI

Select `sb_flop` in Logical Hierarchy. Choose Schematic, then New Schematic View. The first view is the module boundary with its ports. Double-click the module box to expand the mapped contents.



The expanded mapped register schematic in Design Vision.

1. Find `clk`, `rst`, `d`, and `q` at the boundary. Confirm the intended directions and compare them with the RTL declaration.
2. Find the mapped sequential instance `q_reg`. Trace its output to `q` and inspect how reset and data connect through the mapped implementation. Use actual pin names and library semantics when interpreting the cell.
3. Choose View, Zoom, Zoom Fit All to recover the complete view. Zoom into a cell or net when reading labels. The small pane control at the upper right can collapse a pane; double-click its bottom tab to restore it.
4. Select a cell or net to distinguish it from surrounding logic. Use the Schematic menu's fanin and fanout commands when following a larger design. Keep the selected object and direction of tracing explicit.
5. Compare the schematic with `mapped.v` and `area.rpt`. Synthesis may implement the RTL using library-specific cells, constants, or transformations. A familiar-looking drawing is not an equivalence proof.

For larger blocks, inspect register boundaries, muxing, arithmetic widths, and unexpected constant or disconnected signals. An unexpectedly small design can mean intended logic was optimized away because outputs were unused or constraints and connectivity were wrong.

For readability, Design Vision exposes font controls under View, Preferences, Style Settings. Change Normal for labels and Monospace for report text. Prefer enlarging the relevant pane and text before taking a screenshot. Keep the saved source and reports with the image so a reviewer can inspect the underlying evidence.

Connect a timing report to the source

In Design Vision, choose Timing, then Report Timing Path. Start with Delay type `max`, Path type `full`, one worst path per endpoint, and one path per group. Leave From, Through, and To empty for an initial overall report and select To report viewer. Click OK.

Read the startpoint, endpoint, clock, path group, arrival time, required time, and slack. Use the report's object links and the schematic to locate the relevant cell or net. For a focused investigation, return to the dialog and select the intended startpoint or endpoint. Confirm whether the report is analyzing maximum delay or minimum delay before drawing a conclusion.

The same inspection is available in the tool console:

```
report_timing -delay_type max -max_paths 5
report_timing -delay_type min -max_paths 5
report_constraint -all_violators
report_area
```

Positive slack on one reported path does not prove that every path is constrained or that physical timing will pass. Give the PD team the source manifest and timing intent, not just a favorable screenshot.

Prepare a useful RTL handoff

Handoff item	What to include
Source manifest	Revision, top, ordered sources, includes, packages, and parameters.
Interface contract	Ports, widths, clocks, reset behavior, transactions, latency, and backpressure.
Implementation status	Identify real RTL, stubs, black boxes, and unresolved behavior.
Verification evidence	Commands, checks, known limitations, logs, and a representative waveform.
Synthesis evidence	Tool and library, constraints, diagnostics, mapped outputs, area, and timing.

Complete one edit, rebuild, simulation inspection, synthesis check, and schematic inspection on your block before reporting that its development loop works. Keep Compute 1 and Compute 2 deliverables independent; use the architecture interfaces as the shared reference. Close Verdi and Design Vision when finished, then leave the container and log out of the CAE desktop.

Setup reference: <https://kb.wisc.edu/cae-software-guide>

Debug reference: <https://www.synopsys.com/verification/debug/verdi.html>